

# REBATES MANAGEMENT API POC - H2 SETUP AND TESTING RUNBOOK

Version: 1.0

Date: 2026-02-12

Project: rebates-management (Spring Boot)

## 1) PREREQUISITES

- Java 17 installed
- Maven installed (or run from IDE)
- Port 8080 available
- Postman and/or curl available

## 2) START THE APPLICATION

Command:

```
mvn spring-boot:run
```

Expected startup indicators in logs:

- "Started RebatesManagementApplication"
- H2 console enabled at /h2-console
- No datasource initialization errors

## 3) OPEN H2 CONSOLE AND CONNECT

URL:

```
http://localhost:8080/h2-console
```

Connection values:

- JDBC URL: jdbc:h2:mem:rebatedb;MODE=PostgreSQL;DB\_CLOSE\_DELAY=-1;DB\_CLOSE\_ON\_EXIT=FALSE
- User Name: sa
- Password: (empty)

Expected result:

- Connection opens successfully
- Left pane shows schemas/tables

## 4) VERIFY DATABASE CONNECTION

Run SQL:

```
SELECT 1;
```

Expected result:

- Single row containing value 1

## 5) VERIFY TABLE EXISTS

Run SQL:

```
SELECT TABLE_NAME
FROM INFORMATION_SCHEMA.TABLES
WHERE TABLE_NAME = 'REBATE_APPLICATION_AUDIT';
```

Expected result:

- One row with TABLE\_NAME = REBATE\_APPLICATION\_AUDIT

## 6) VERIFY STORED PROCEDURE/FUNCTION ALIAS EXISTS

Run SQL:

```
SELECT ALIAS_NAME
FROM INFORMATION_SCHEMA.FUNCTION_ALIASES
WHERE ALIAS_NAME = 'SP_SUBMIT_REBATE_APPLICATION';
```

Expected result:

- One row with ALIAS\_NAME = SP\_SUBMIT\_REBATE\_APPLICATION

## 7) API CHECK #1 - VERIFY DB ARTIFACTS

curl command:

```
curl --location --request GET 'http://localhost:8080/api/v1/rebates/db/artifac
```

ts'

Expected HTTP status:

- 200 OK

Expected response body:

```
{
  "auditTableExists": true,
  "rebateProcedureExists": true,
  "databaseProduct": "H2",
  "message": "H2 artifacts are available"
}
```

## 8) API CHECK #2 - SUCCESS SCENARIO (SUBMIT REBATE)

curl command:

```
curl --location 'http://localhost:8080/api/v1/rebates/applications' \
--header 'Content-Type: application/json' \
--data '{
  "vendorId": "VEND-1001",
  "vendorName": "Acme Supplies",
  "productCode": "PRD-REB-01",
  "quantity": 25,
  "unitPrice": 150.75,
  "rebatePercentage": 12.5,
  "remarks": "Quarterly rebate request"
}'
```

Expected HTTP status:

- 201 Created

Expected response body pattern:

```
{
  "transactionId": "<uuid>",
  "applicationId": <number>,
  "rebateAmount": 470.00,
  "status": "SUCCESS",
  "message": "Rebate application submitted successfully",
  "processedAt": "<timestamp>"
}
```

Post-call DB verification SQL:

```
SELECT audit_id, vendor_id, vendor_name, product_code, quantity,
       unit_price, rebate_percentage, remarks, created_at
FROM rebate_application_audit
ORDER BY created_at DESC;
```

Expected result:

- New row inserted for the API call

## 9) NEGATIVE SCENARIOS

A) Missing mandatory field (vendorId)

curl command:

```
curl --location 'http://localhost:8080/api/v1/rebates/applications' \
--header 'Content-Type: application/json' \
--data '{
  "vendorName": "Acme Supplies",
  "productCode": "PRD-REB-01",
  "quantity": 25,
  "unitPrice": 150.75,
  "rebatePercentage": 12.5
}'
```

Expected HTTP status:

- 400 Bad Request

Expected response body pattern:

```
{
  "errorCode": "VALIDATION_ERROR",
  "errorMessage": "Request validation failed",
  "details": ["vendorId: vendorId is required"],
  "timestamp": "<timestamp>"
}
```

B) Business rule violation (rebatePercentage > 50)

curl command:

```
curl --location 'http://localhost:8080/api/v1/rebates/applications' \
--header 'Content-Type: application/json' \
--data '{
  "vendorId": "VEND-1001",
  "vendorName": "Acme Supplies",
  "productCode": "PRD-REB-01",
  "quantity": 25,
  "unitPrice": 150.75,
  "rebatePercentage": 55,
  "remarks": "Invalid percentage"
}'
```

Expected HTTP status:

- 422 Unprocessable Entity

Expected response body pattern:

```
{
  "errorCode": "BUSINESS_ERROR",
  "errorMessage": "rebatePercentage cannot exceed 50",
  "details": [],
  "timestamp": "<timestamp>"
}
```

C) Invalid numeric value (quantity <= 0)

curl command:

```
curl --location 'http://localhost:8080/api/v1/rebates/applications' \
--header 'Content-Type: application/json' \
--data '{
  "vendorId": "VEND-1001",
  "vendorName": "Acme Supplies",
  "productCode": "PRD-REB-01",
  "quantity": 0,
  "unitPrice": 150.75,
  "rebatePercentage": 12.5,
  "remarks": "Invalid quantity"
}'
```

Expected HTTP status:

- 400 Bad Request

Expected response body pattern:

```
{
  "errorCode": "VALIDATION_ERROR",
  "errorMessage": "Request validation failed",
  "details": ["quantity: quantity must be greater than 0"],
  "timestamp": "<timestamp>"
}
```

10) TROUBLESHOOTING QUICK CHECKS

- If /h2-console does not open, verify app started on port 8080.
- If function alias is missing, verify schema.sql ran during startup.
- If POST fails with 500, inspect app logs for database exception details.
- If curl cannot connect, verify URL and local firewall/port usage.

END OF DOCUMENT